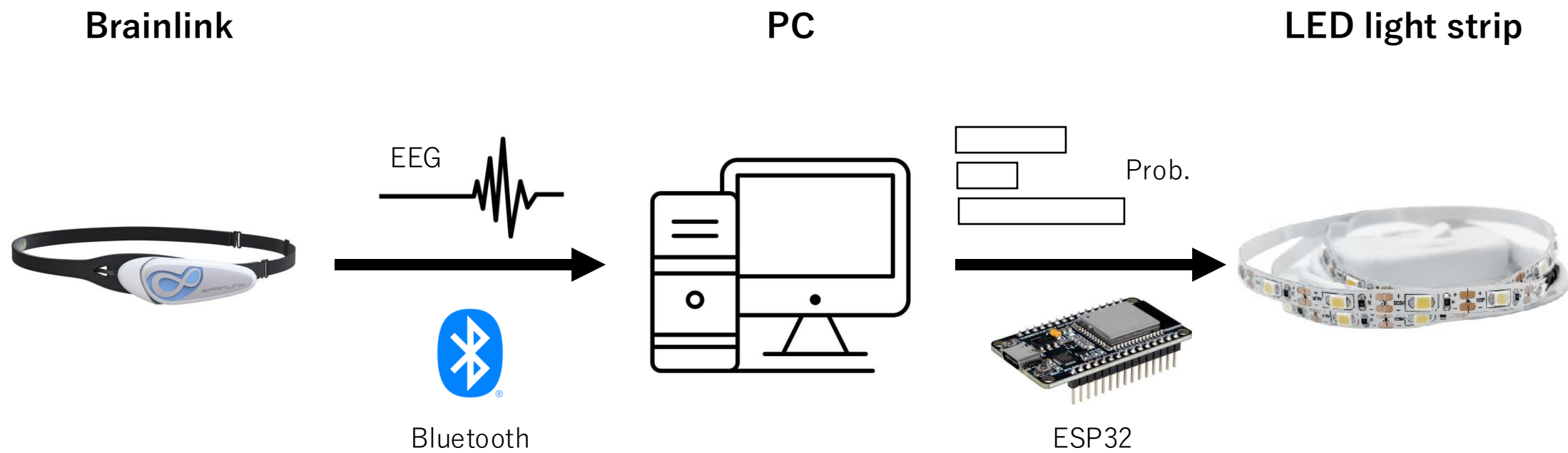


114-2 電工實驗（生醫工程）期末專題成果報告

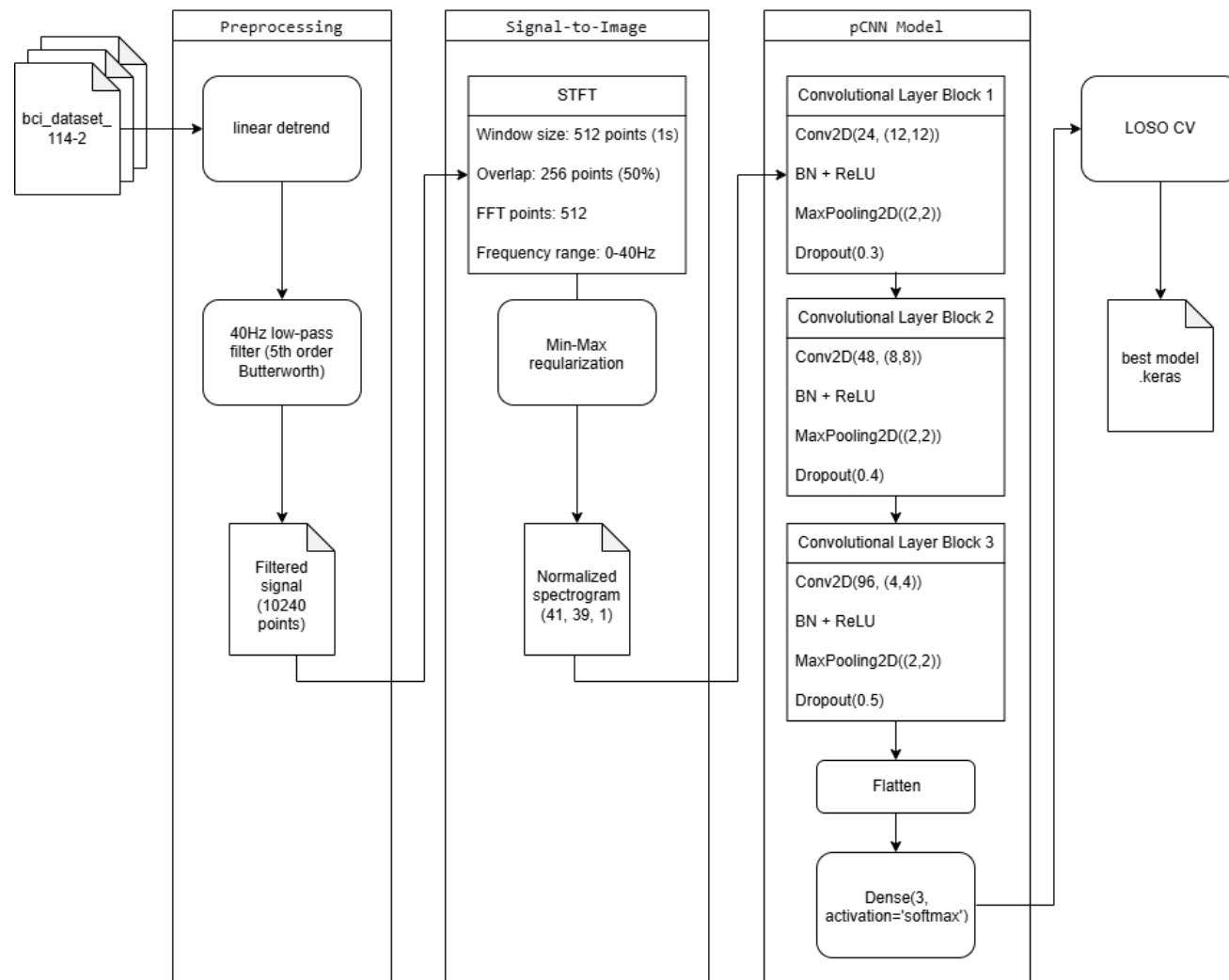
基於腦波狀態辨識的智能氛圍燈系統

張博翔、劉漾晴

研究目標



Task 1 – EEG 分類

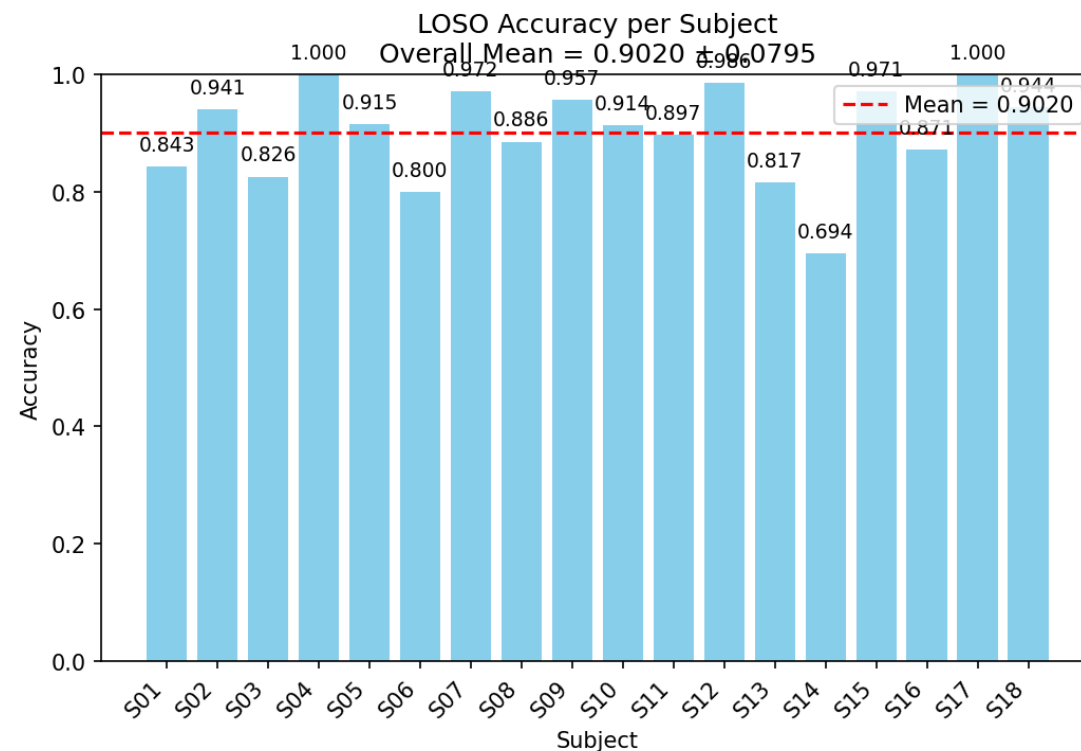


EEG 分類模型訓練流程圖

Leave-One-Subject-Out (LOSO) Cross-Validation

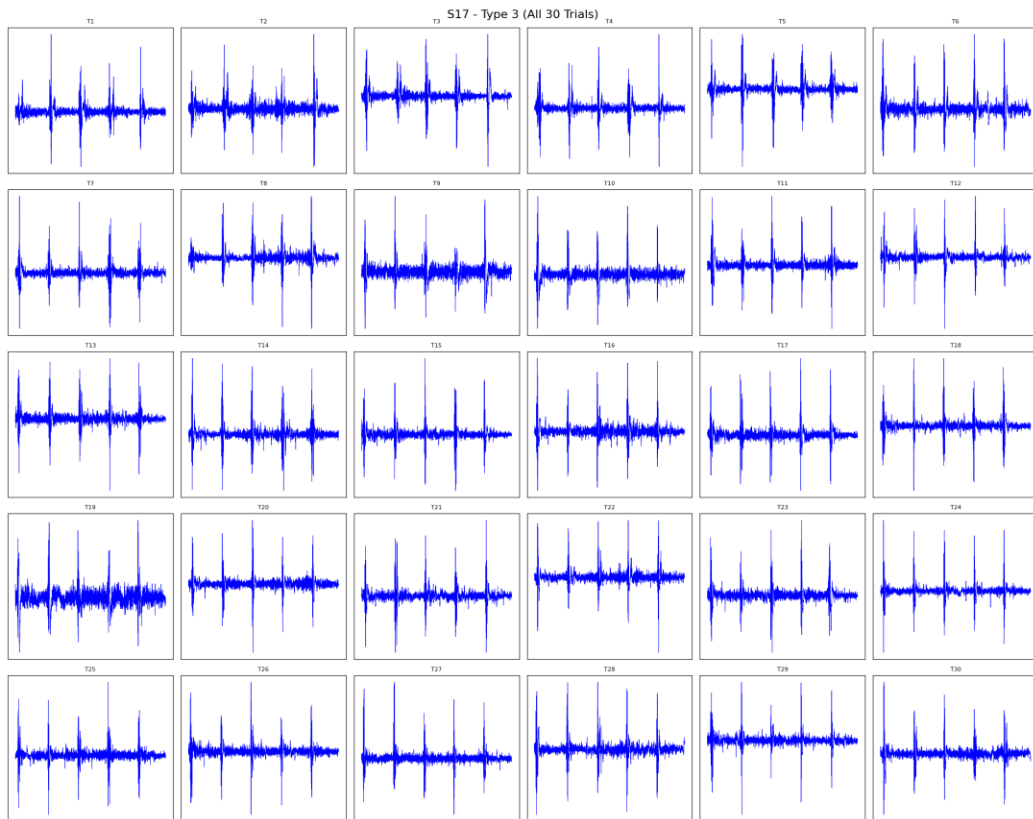


20 秒模型 (Relax / Focus / Blink)

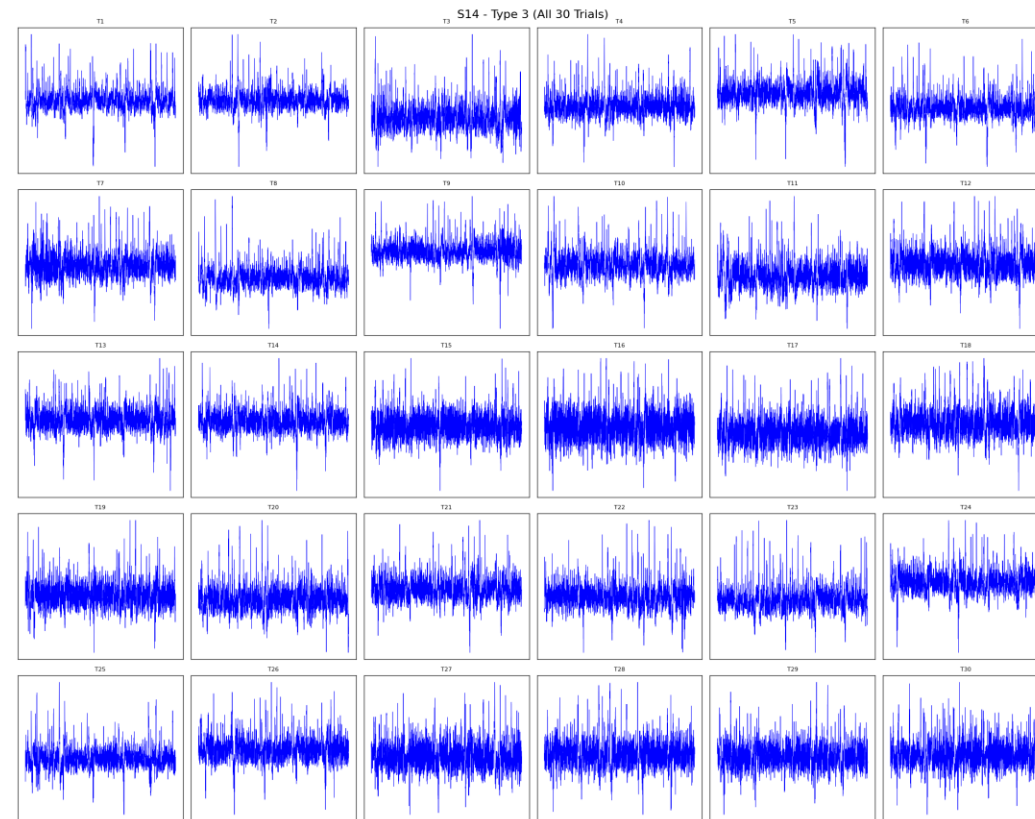


20 秒模型 (Relax / Focus / Blink)

眨眼類別優化 – 4 秒獨立模型

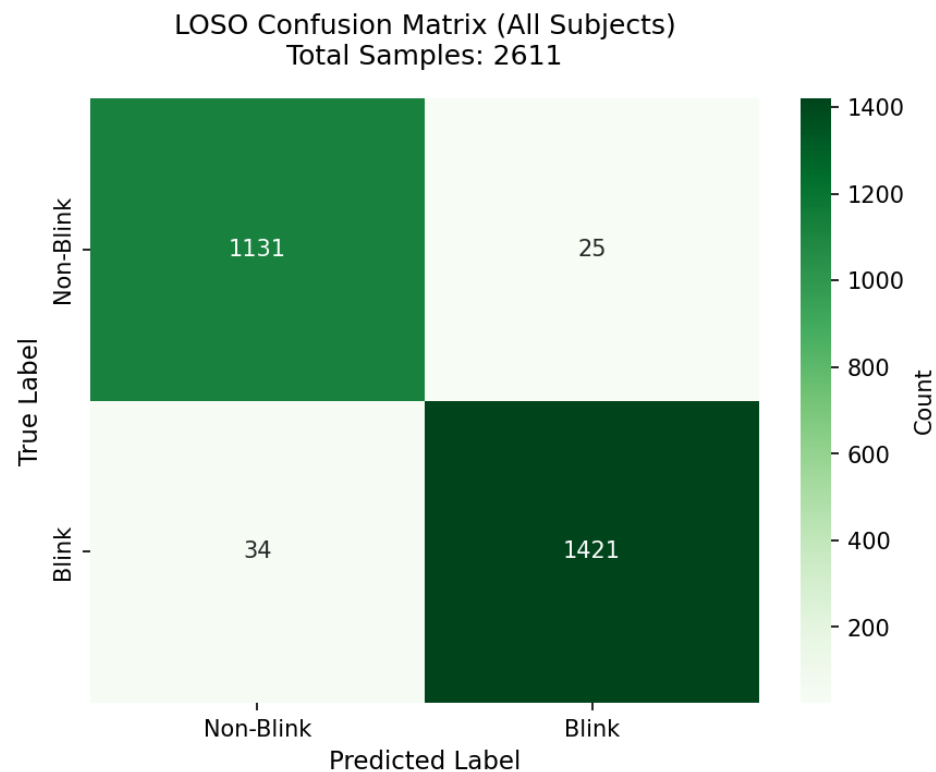


S17 Blink – raw data

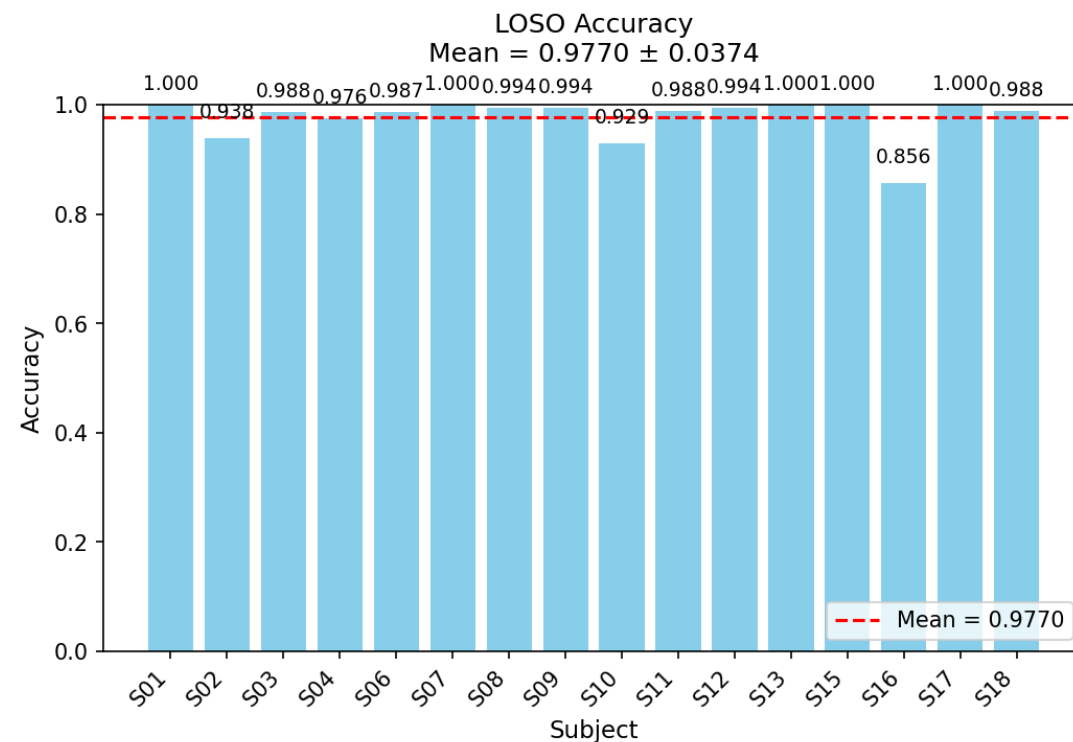


S14 Blink – raw data (剔除)

LOSO Cross-Validation – 4 秒獨立模型

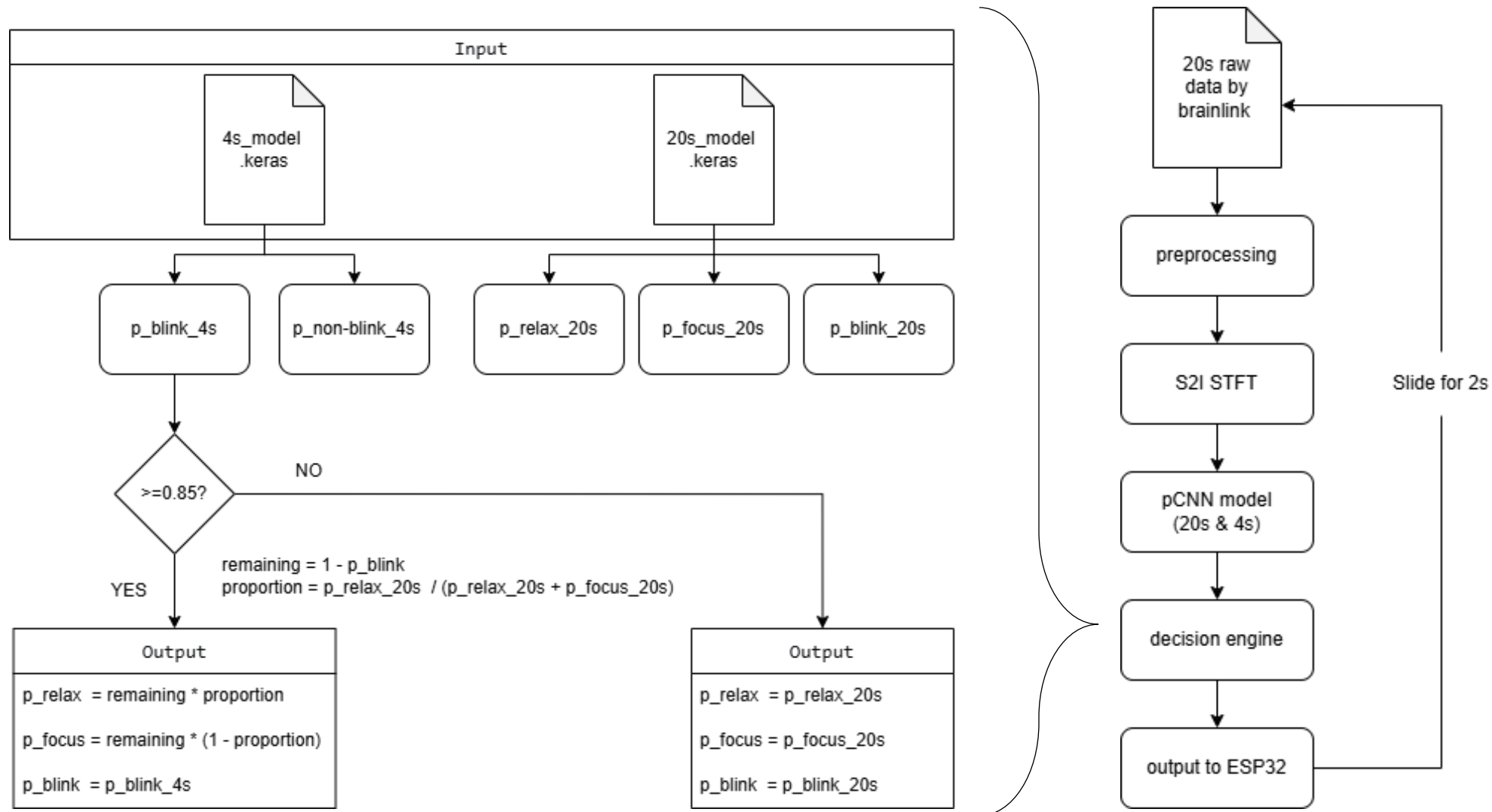


4 秒模型 (Non-Blink / Blink)



4 秒模型 (Non-Blink / Blink)

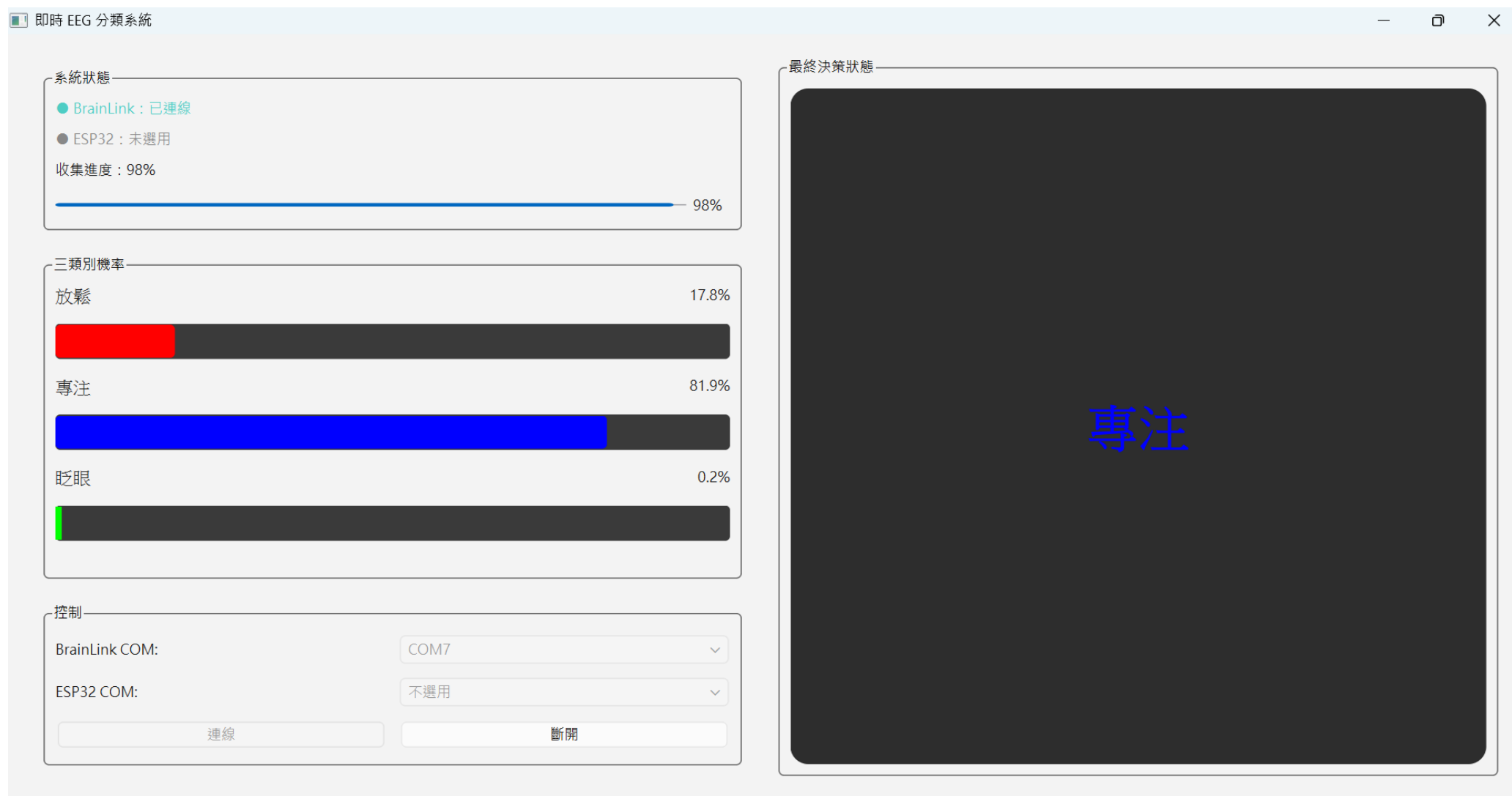
及時決策邏輯



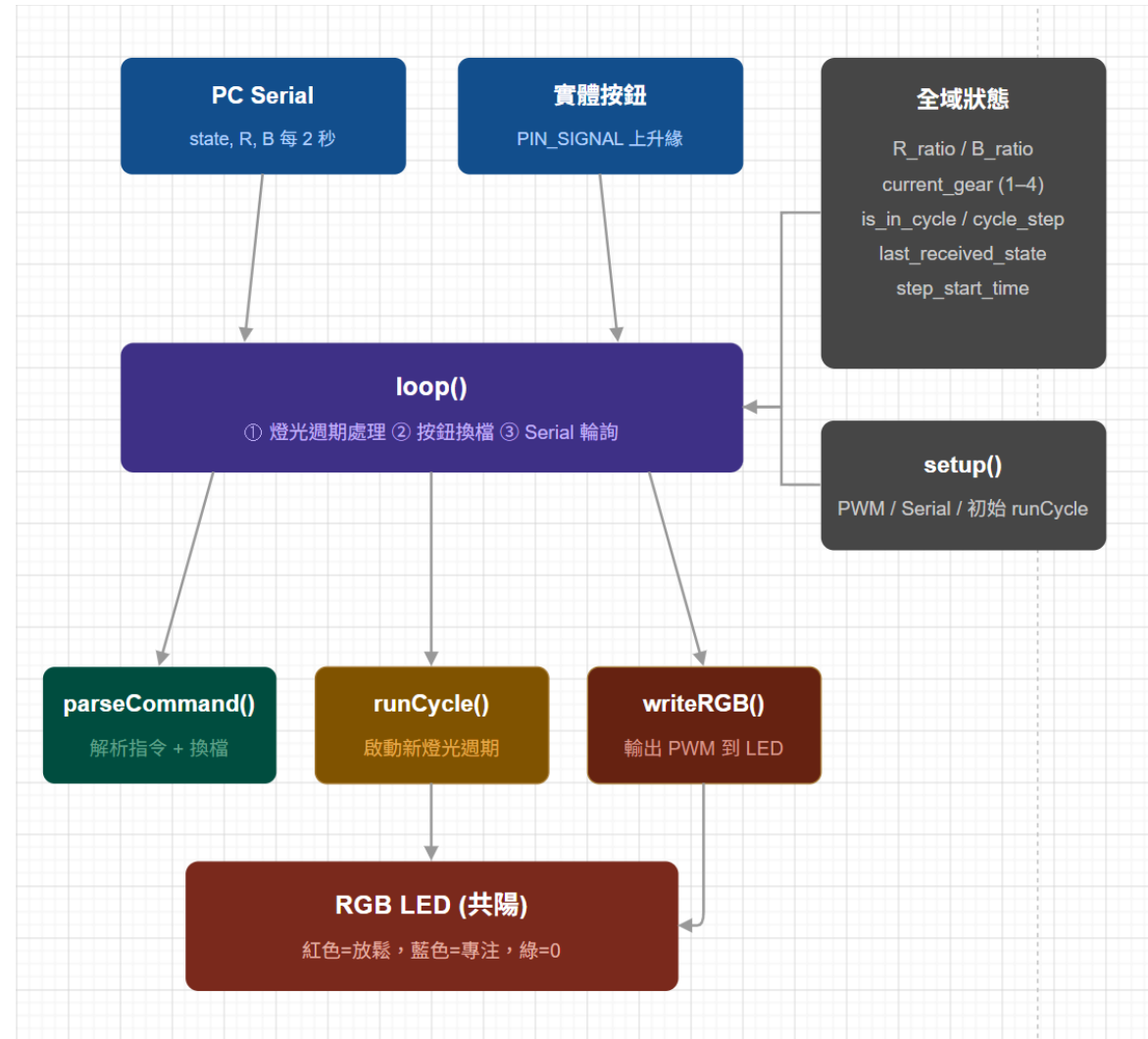
狀態機率決策邏輯流程圖 (decision engine)

及時決策流程圖

GUI 程式介面展示

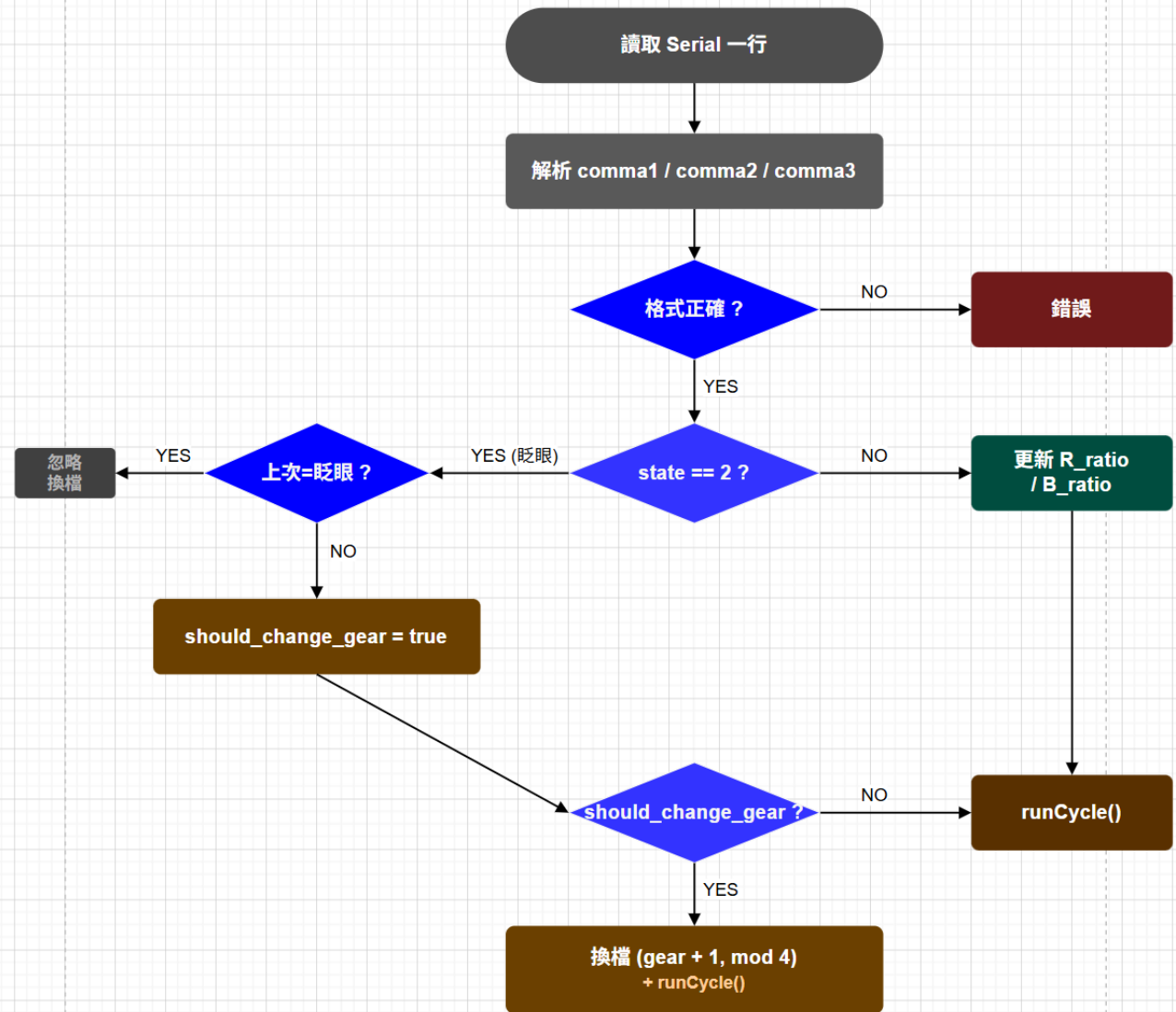


Task 2 – Part1 氛圍燈驅動



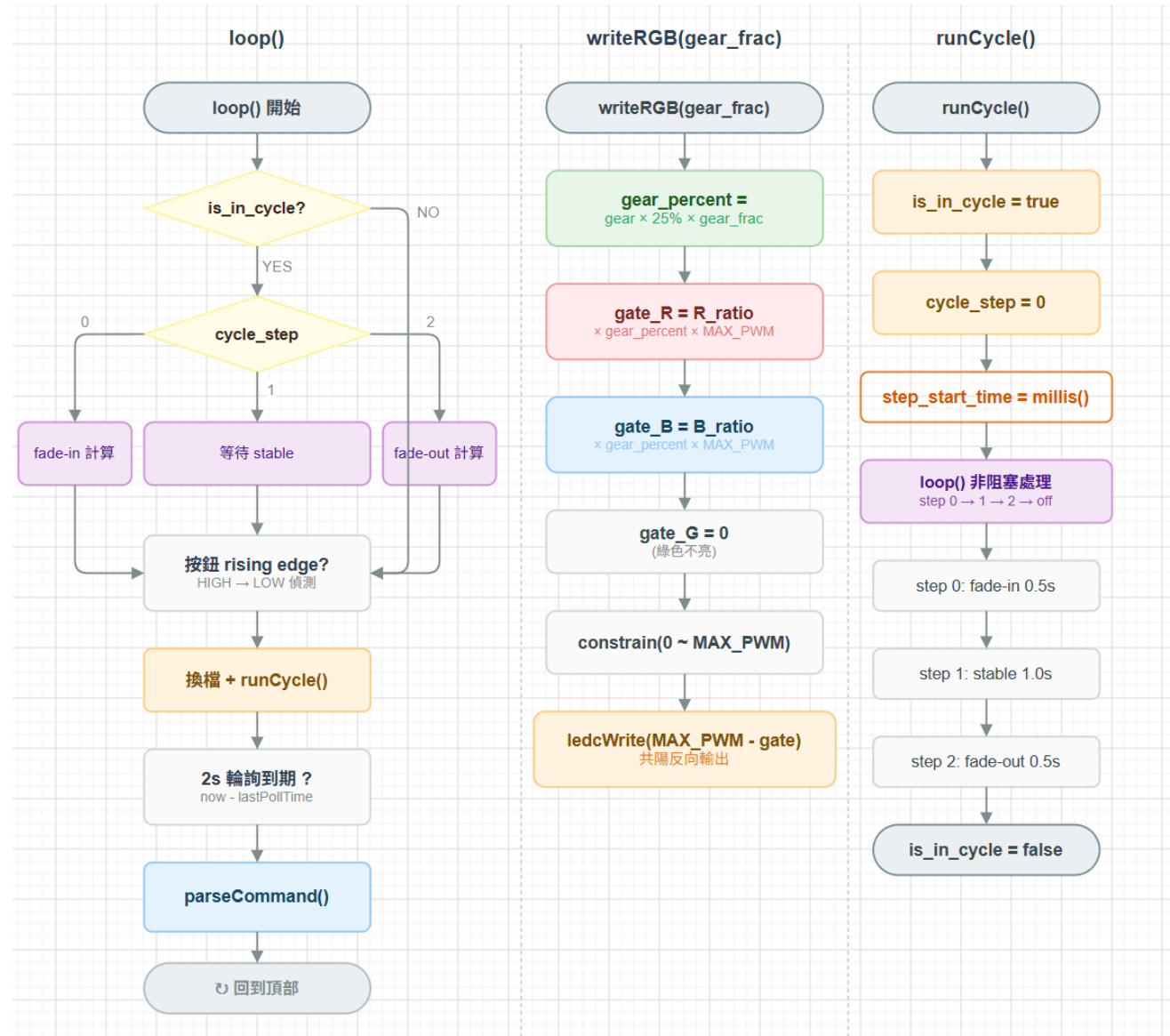
MCU Drive Workflow

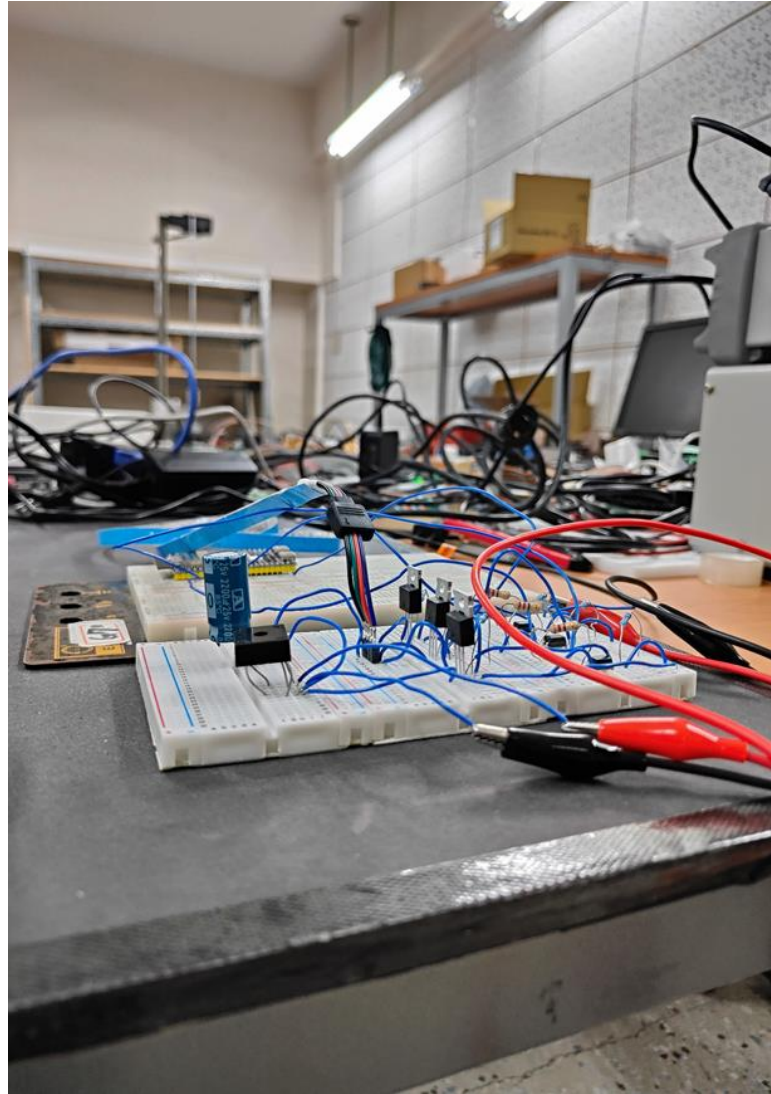
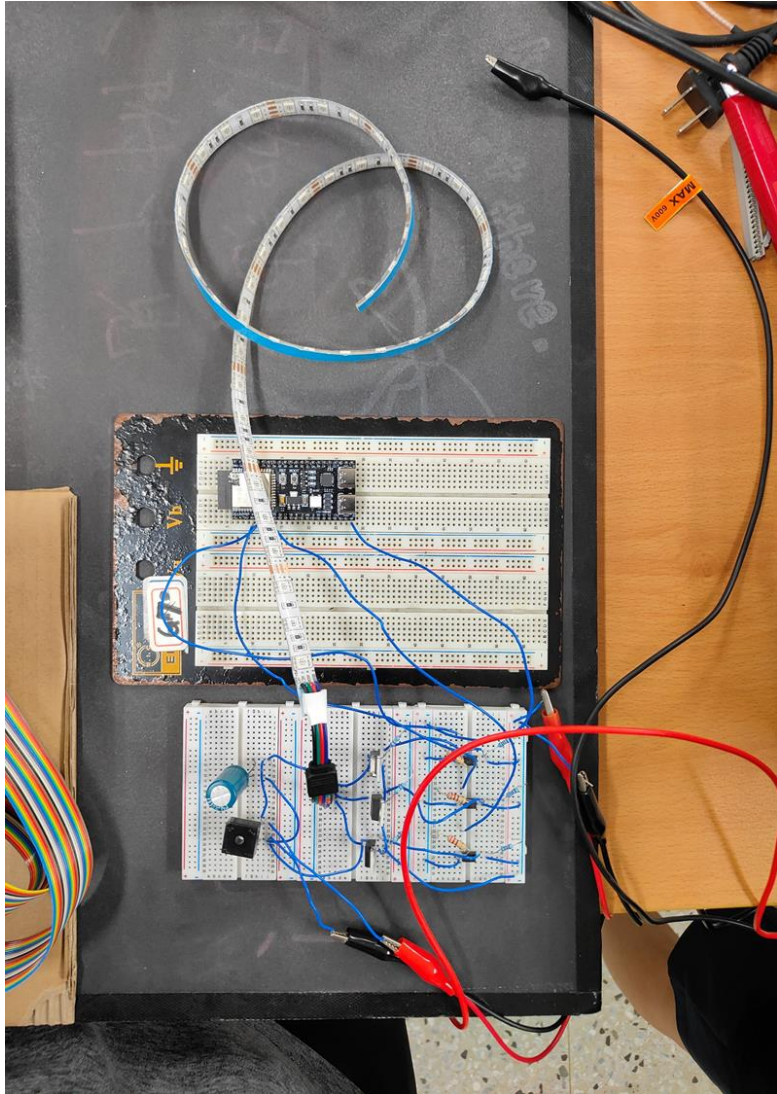
- parseCommand



MCU Drive Workflow

- loop
- runCycle
- writeRGB





Task 2 – Part2 氛圍燈電路

Hardware Circuit Design

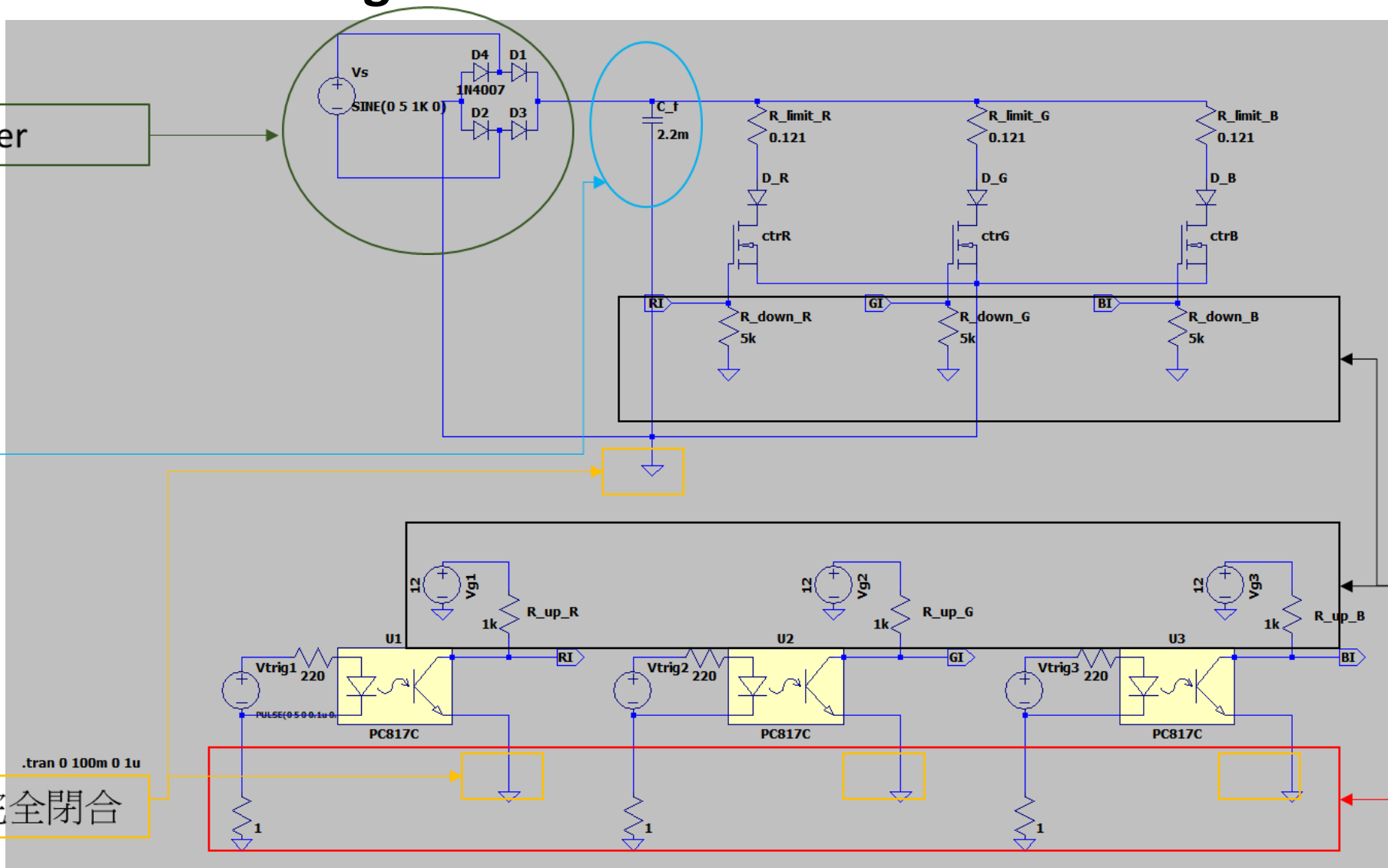
Source 跟 rectifier

$$C = \frac{I}{2f \cdot dv}$$

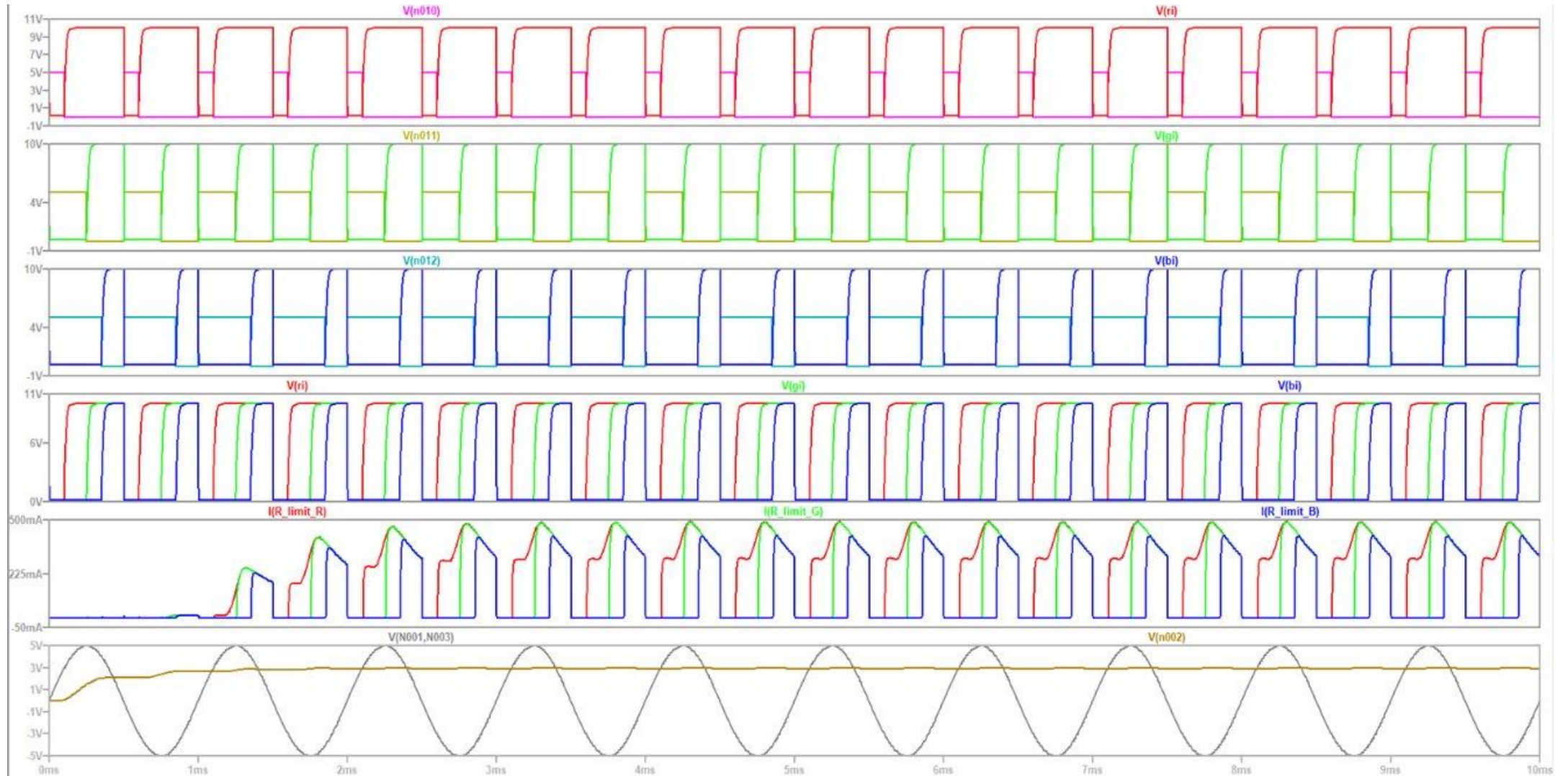
確定Gate可以完全閉合

上/下拉電阻
穩定電位

PC / 系統隔離



Simulation and Verification



DEMO 影片

系統狀態

- BrainLink : 未連線
- ESP32 : 未選用

收集進度 : 0%

0%

三類別機率

放鬆 33.0%

專注 33.0%

眨眼 34.0%

控制

BrainLink COM: COM7

ESP32 COM: COM6

連線 斷開

最終決策狀態

等待中



DEMO 影片_亮度控制

ch.ino diagram.json Library Manager

```
/*
 * ESP32_Driver_Modified.ino
 * 驅動 RGB LED：紅藍亮度固定 50%，綠色不亮
 * 換檔方式：實體按鈕 或 透過 Serial 輸入 'G'
 */

#include <math.h>

// ===== 可調參數區 =====
const int PIN_R      = 4; // R 腳位 (紅色)
const int PIN_G      = 5; // G 腳位 (綠色 - 不使用，設為 0)
const int PIN_B      = 6; // B 腳位 (藍色)
const int PIN_SIGNAL = 0; // 實體按鈕腳位

const int PWM_FREQ   = 2000; // PWM 頻率 Hz
const int PWM_RES     = 10;  // PWM 解析度 bit
const int MAX_PWM     = 1023; // 最大 PWM 值 (2^10 - 1)

const int POLL_INTERVAL_MS = 100; // 輪詢間隔 ms (縮短以加快 Serial 反應)

// 燈光週期參數
const float FADE_IN_SEC   = 0.5; // 緩啟動時間 (秒)
const float STABLE_SEC    = 1.0; // 穩定維持時間 (秒)
const float FADE_OUT_SEC  = 0.5; // 緩關閉時間 (秒)
const int FADE_STEPS      = 20;  // 漸變步數

// 檔位參數
const int MAX_GEAR = 4;
const unsigned long GEAR_COOLDOWN_MS = 500; // 換檔冷卻 0.5 秒
// =====

int current_gear = 1;

// 固定 R/B 比例為 50%
float R_ratio = 0.50;
float B_ratio = 0.50;

unsigned long lastGearChangeTime = 0;
unsigned long lastPollTime = 0;
unsigned long step_start_time = 0;
bool is_in_cycle = false;
```

Simulation

參考資料

- [1] Z. Tayeb, J. Fedjaev, N. Ghaboosi, C. Richter, L. Everding, X. Qu, Y. Wu, G. Cheng, and J. Conradt, “Validating deep neural networks for online decoding of motor imagery movements from eeg signals,” *Sensors*, vol. 19, p. 210, Jan. 2019.
- [2] P. Bizopoulos, G. I. Lambrou, and D. Koutsouris, “Signal2image modules in deep neural networks for eeg classification,” *Proceedings of the 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 702–705, 2019.